

FIT3152 Data analytics – 2026: Assignment 3

Your task	- The objective of this assignment is to gain familiarity with Natural Language Processing and network analysis using R. - This is an individual assignment.
Value	- This assignment is worth 25% of your total marks for the unit. - It has 32 marks in total.
Suggested Length	8 – 10 A4 pages, approximately 1,000 words, (for your report) + extra pages as appendix for your R script. - Font size 11 or 12pt, single spacing.
Due Date	11.55pm Friday 12th June 2026
Submission	You will submit 3 files: - Submit your report as a single pdf file. - Submit your video file as an mp4, m4v etc. - Submit your corpus as either a zipped folder or csv file on Moodle. <i>Please do not put your pdf and video in your zip file.</i> - Use the naming convention: <i>FirstnameSecondnameID.{pdf, zip, csv, mp4}</i> - Turnitin will be used for similarity checking all written submissions.
Generative AI Use	AI & Generative AI tools MUST NOT BE USED within this assessment / task for the following reasons: This whole assessment task requires students to demonstrate human knowledge and skill acquisition without the assistance of AI.
Late Penalties	5% (1.6 mark) deduction per calendar day for up to one week. - Submissions more than 7 calendar days after the due date will receive a mark of zero (0) and no assessment feedback will be provided.

Instructions and data

In this assignment, you will create a corpus of documents and analyse the relationships between them, as well as the relationships between the important words used in these documents.

Background material for this assignment is covered in Weeks 9, 10 and 11. You are free to consult any other references, including those listed at the end of the document.

There are two options for compiling your written report:

- (1) You can create your report using any word processor with your R code pasted in as machine-readable text as an appendix, and save as a pdf, or
- (2) As an R Markdown document that contains the R code with the discussion/text interleaved. Render this as an HTML file and save as a pdf.

Your video report should be approximately 5 minutes in length and less than 100MB in size. You may need to reduce the resolution of your original recording to achieve this. Use a standard file format such as .mp4, or mov for submission.

Tasks

1. Collect a set of (machine-readable text) documents. There is no restriction on the type of material you can choose although please avoid texts that might be offensive to people. As a guide, you should aim for the following:
 - Each document should be at least 200 words in length. Collect at least 20 documents.
 - You should have at least 3 different genres (for example, fiction, news, film reviews, sport, politics) in your collection of documents. Label each document as belonging to one of the genres in your corpus.
 - You can collect the documents as pdfs or as copied text from web-based articles or other files.
 - Reference the sources of your documents (URL or bibliographic citation: APA or Harvard style). **(1 Mark)**
2. Create your corpus using one of the methods covered in lectures or applied sessions. This could either be a folder of text files or a suitably formatted CSV file. Use suitable identifiers for your text file names or document IDs so that you can recognize the document in your clustering or network graphs.
 - The type of original material you collect will determine the way you need to do this. For some formats you can simply copy and paste the text into an empty text file. For other formats you may find it simpler to use “export”, or “save as” function in software. Describe the process you follow for this step in your report. **(3 Marks)**
3. Follow the text processing steps covered in lectures and applied sessions to create your Document-Term Matrix (DTM).
 - As part of this process, you may need to make particular text transformations to preserve key words, or remove unwanted terms, unwanted characters or artefacts from the original formatting. Describe this processing in your report or state why you did not need to do so.
 - Your DTM should contain approximately 25 tokens after you have removed sparse terms. You will need to do this by trial-and-error to get the right number of tokens.
 - Include your DTM as a table in the appendix of your report. **(3 Marks)**
4. Create a hierarchical clustering of your corpus and show this as a dendrogram.
 - Use the cosine distance between each document for clustering.
 - Identify which cluster each document belongs to.
 - Calculate the accuracy of the clustering by genre.
 - Give a qualitative description of the quality of the clustering using examples from the texts where appropriate. **(4 Marks)**

- 5 Analyse the sentiment of each document in your corpus using one of the methods covered in Week 11.
- Describe how you performed your sentiment analysis, and dictionary used.
 - Is there a difference in sentiment between genres, for example average level or variability? Are these differences significant? **(3 Marks)**
- 6 Create a single-mode network showing the connections between the documents based on the number of shared terms.
- To do this you will need to first calculate the connections between each document using the method shown in Week 11, or another method of your choice.
 - What does this graph tell you about the relationship between the documents? Are there any groups in the data you can clearly identify? What are the most important (central) documents in the network?
 - Improve your graph over the basic example given in Week 11 to highlight as many interesting features of the data you can. For example, the strength of connections, the relative importance of nodes, sentiment, genres, groups. **(3 Marks)**
- 7 Repeat all the activities in Question 6, but now looking at the words (tokens). **(3 Marks)**
- 8 Create a bipartite (two-mode) network of your corpus, with document ID as one type of node and tokens as the other type of node.
- To do this you will need to transform your data into a suitable format.
 - What does this graph tell you about the relationship between words and documents? Are there any groups in the data you can clearly identify?
 - Improve your graph over the basic example given in Week 11 to highlight as many interesting features of the data you can. For example, the strength of connections, the relative importance of nodes, sentiment, genres, groups. **(3 Marks)**
- 9 Write a brief report (suggested length 8 – 10 pages).
- Briefly summarise your results identifying important documents, tokens and groups within the corpus. Comment on the relative effectiveness of clustering versus network analysis to identify important groups and relationships in the data.
 - Investigating methods not included in lectures or lecture notes, can you suggest improvements to text processing used in this assignment to better discriminate between the documents studied? Describe briefly how and why these methods work. To do this you may want to refer to recent references on Natural Language Processing. You are not required to implement these improvements.
 - Include your R script as an appendix. Use commenting in your R script, where appropriate, to help a reader understand your code. Alternatively combine working, comments and reporting in R Markdown. **(5 Marks)**
- 10 Record a short presentation using your smart phone, Zoom, or similar method. Your face should be visible in the video. Your presentation should be approximately 5 minutes in length and summarise your main findings, as well as describing how you conducted your research and any assumptions made. **(Submission Hurdle and 4 Marks)**

Software

It is expected that you will use R for your data analysis and graphics and tables. You are free to use any R packages you need but please list these in your report and include in your R code.

References

Statistical Analysis of Network Data with R, Kolaczyk, E. D., Csárdi, G. Springer 2020. Chapters 1 – 4

A User's Guide to Network Analysis in R, Luke, D. A. Springer 2015.

Network visualization with R, PolNet 2018 Workshop <https://kateto.net/>

Bipartite/Two-Mode Networks in igraph, Phil Murphy & Brendan Knapp. <https://rpubs.com/tm> and iGraph package manuals.

Text Data Mining, Chengqing Zong, C., Xia, R., Zhang, J., Springer Nature, Singapore, 2021.